



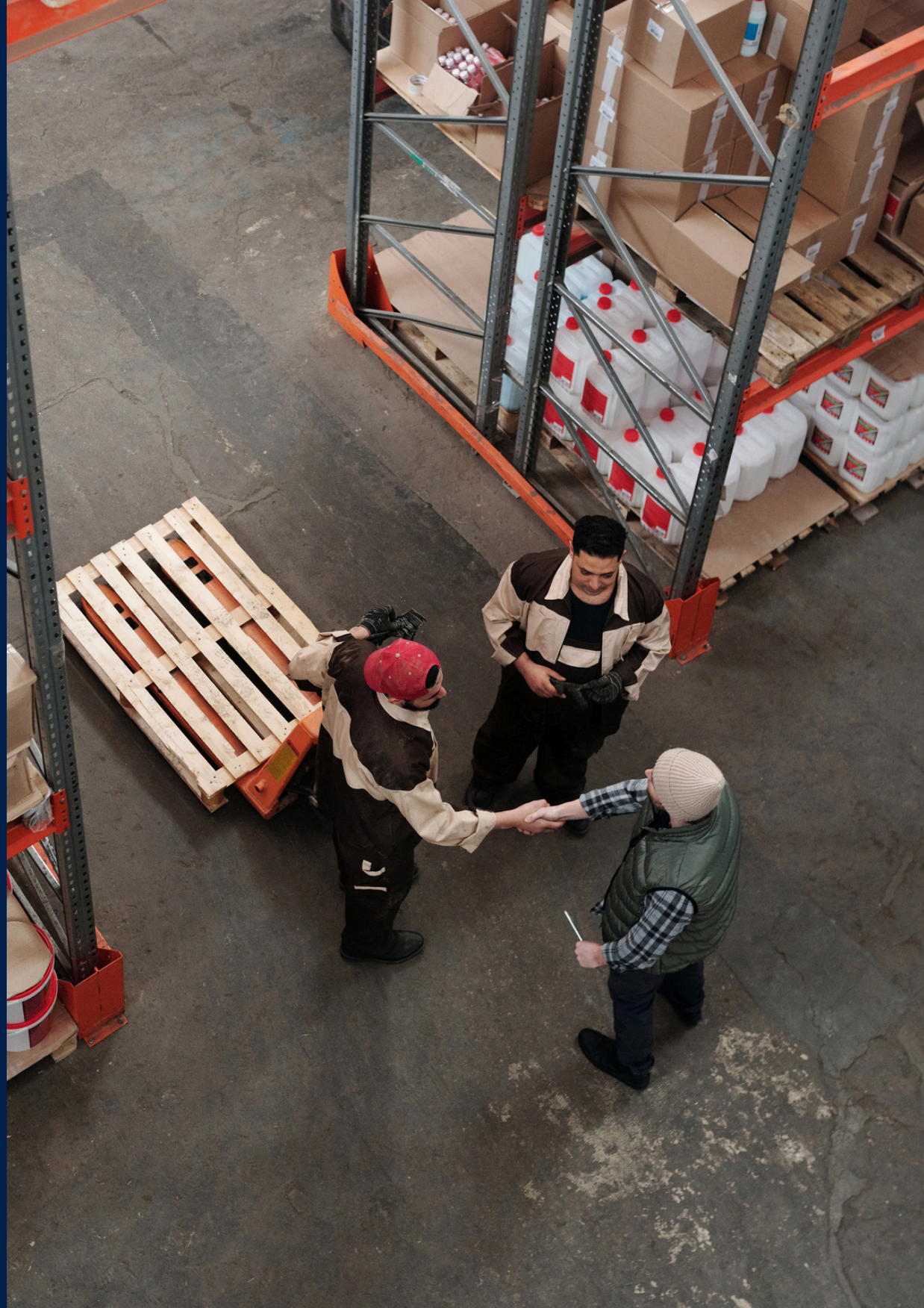
CASE STUDY

TEXTILE COMPANY: DRIVING OPERATIONAL EFFICIENCY

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Problem Statement: To Increase The Yield Of Production

The textile manufacturing unit, operating 150 autoloom machines with different fabric variants, was facing a persistent issue of yarn breakages occurring per day, especially during dry seasonal months. This resulted in loss in daily production output. Despite regular maintenance , the root cause remained unidentified. The factory team lacked a structured diagnostic approach leading to lower yield of production of fabric .





Company Overview

Shree RoopRajat Weaving Mills LLP Textiles is a B2B textile manufacturer

- **produces variety of different weaves like canvas, drill, ducks, twill, plain, oxford, satin, cambric, for garment brands and regional distributors.**
- **Works with a team of 150 autoloom operators and 5 supervisors.**
- **The unit delivers 10,000–50,000 meters of fabric daily, generating annual revenue of ₹1.5–₹2 crore.**

Known for reliability and custom fabric solutions, Shree RoopRajat Weaving Mills LLP Textiles plays a key role in regional supply chains.

Competitors in the market

The textile manufacturing industry is highly competitive, with several established and emerging players offering similar fabric solutions to B2B garment manufacturers.

- **Radhe Krishna Cotweaving**
- **Unique multitex**
- **Multitex**
- **Balaji enterprises**
- **AK india**

Deciphering The Problem

1. Recurring Yarn Breakages

- Looms experienced yarn breakages per day, causing frequent production halts.
- This led to inconsistent output and delivery disruptions, especially in dry seasons.

4. Inferences from Operations Data

- Data showed higher breakage rates during day shifts in dry months.
- Shifts with better humidifier practices had lower yarn snapping incidents.

2. Gap in Standard Operating Procedures

- SOPs lacked environmental guidelines —no mandate on maintaining optimal humidity.
- Operators often turned off humidifiers to reduce excess humidity, unaware of its impact.

5. Business Impact and Responsibility

- Production delays sometimes affected order fulfillment timelines.
- There is a significant loss in output along with substantial monthly losses due to inefficiencies and material waste.
- .Therefore investigating the root cause, and implementing a sustainable solution was required.

3. Diagnostic Gaps Despite Equipment Reliability

- The machines were routinely maintained and mechanically sound, eliminating equipment failure as a direct cause of yarn breakages.
- The factory used both high-quality and lower-quality yarns, yet applied identical chemical treatments (used to enhance CSP – Count Strength Product) across all batches, without adjusting for material variability.
- This one-size-fits-all approach may have contributed to inconsistent yarn behavior during weaving, especially under changing ambient conditions.

MEASUREMENT:

The measurement required met the width specifications .

MAN (HUMAN FACTORS):

When humidifier was manually switched off there was an increase in yarn breakage frequency during those shifts.

MACHINE (EQUIPMENT CONDITION):

SOPs lacked clear guidance on environmental controls like temperature or humidity which led to inability to trace or prevent recurring breakage incidents.

6M ANALYSIS

MATERIAL (YARN QUALITY & TREATMENT):

The same CSP enhancers was applied to all yarns, regardless of grade which led to increased brittleness in low-grade yarns, especially under dry conditions.

METHOD (PROCESS & SOP GAPS):

All autolooms were regularly maintained and passed mechanical inspections. Machines were not the primary cause of the issue.

MOTHER NATURE (ENVIRONMENT):

Humidity levels during peak dry months often dropped due to which yarn fibers loses flexibility, increasing tension-related snapping.

Product Thinking Approach

1. UNDERSTAND THE “USER” PROBLEM:

User Need: Reliable production with minimal downtime to fulfill B2B orders on time

Pain Point: Frequent yarn breakages , causing delays, low material quality and waste

Challenge: No clarity on root cause — machines were fine, but output was inconsistent

Reframed problem as a system-level issue involving material, environment, and human behavior.

2. INVESTIGATE ROOT CAUSES (HYPOTHESIS-LED DIAGNOSIS):

Hypothesis 1: Breakages spike more during dry month seasons and lower humidity level provided by humidifiers

Hypothesis 2: Uniform chemical treatment for different yarn grades creates brittleness in low-CSP yarn

Hypothesis 3: Manual shutdown of humidifiers during shifts affects yarn performance

3. COLLECT AND ANALYZE DATA:

Reviewed shift logs, humidity levels, yarn batches, and breakage events across 6 weeks.

Key finding:

Breakages increased when RH was below ambient humidity level;

Lower-CSP yarns broke more often when given the same treatment as premium yarn;

Due to turning off humidifier, breakage counts increased.

Solution Implementation

1. Humidifier Usage Standardization

- Updated SOPs to include a mandatory guideline to maintain ambient humidity in the weaving shed.
 - Installed digital hygrometers with real-time humidity monitoring and automated alerts when levels fall below threshold.
 - Introduced a shift-level compliance checklist, ensuring the humidifier remains operational and logged.
 - Conducted awareness sessions for all workers to explain the scientific link between humidity, yarn flexibility, and breakage.
- Result: Yarn breakages reduced by ~60% during low-humidity periods, improving fabric consistency.

2. Material-Specific Chemical Treatment Protocol

- Audited yarn batches and identified significant variation in CSP (Count Strength Product) across suppliers.
 - Introduced two-tier chemical treatment SOPs:
 - Tier 1: For high-CSP yarns (CSP > 2000) – standard chemical soak and drying.
 - Tier 2: For low-CSP yarns (CSP 1600–1900) – reduced soaking time and lower drying temperature to prevent brittleness.
 - Created yarn identification labels and sorting bins to prevent treatment mix-ups at the pre-weaving stage.
- Result: Reduced yarn snapping by ~40% for lower-grade yarns under otherwise stressful environmental conditions.



Quantified Business Impact



- LOOM UPTIME INCREASED
- LED TO A STEADY RISE IN DAILY FABRIC PRODUCTION.
- MONTHLY SAVINGS: ~₹95,000 FROM REDUCED WASTE AND DOWNTIME
- CULTURAL SHIFT: OPERATORS BECAME MORE PROCESS-AWARE AND DATA-CONSCIOUS

KPI	Before Optimization	After Optimization
Loom Uptime	63%	91%
Yarn Breakage Rate	8–10 breaks/shift	<4 breaks/shift
Fabric Output	~ 17,500 meters/day	~27,500 meters/day
Waste Cost	₹1.5lakh/month	~₹55,000/month

Key Product Skills Demonstrated

Root Cause Analysis (5M Framework)

Used the 6M Analysis (Man, Machine, Material, Method, Mother Nature)

Hypothesis-driven Experimentation

Collected data (breakage counts, humidity levels), validated findings, and implemented targeted solutions.

KPI-Focused Operations Optimization

Improved daily operations by tracking and optimizing key metrics.

Cross-functional team coordination

Collaborated across departments and worker levels to implement change. Ensured SOP changes were understood and adopted.

Systems thinking and continuous improvement

Viewed the factory as a connected system, where environment, process, and people interact.



Thank You.